Comparison of surds:

1. Increasing Power $\sqrt[3]{25}$ and $\sqrt{8}$
2. Decreasing Power 5^{54} and 11^{36}
3. Squaring both sides $6+\sqrt{5}$ and $5+\sqrt{11}$
4. Observing pattern

Must memorize:

- $\sqrt{2}=1.414$
- $\sqrt{3}=1.732$
- $\sqrt{5}=2.236$

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① Which of the following is greatest?

निम्न से सबसे बड़ा:

LCM(2,3,4,6)=12

a) $\sqrt{2}$ b) $\sqrt[3]{3}$ c) $\sqrt[4]{4}$ d) $\sqrt[6]{6}$

$2^{\frac{1}{2}\times 12}$ $3^{\frac{1}{3}\times 12}$ $4^{\frac{1}{4}\times 12}$ $6^{\frac{1}{6}\times 12}$

64, 81, 64, 36

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(2) Find the smallest among $2\frac{1}{12}, 3\frac{1}{18}, 4\frac{1}{24}, 6\frac{1}{36}, 12\frac{1}{72}$ is

72 ×

$2\frac{1}{12}, 3\frac{1}{18}, 4\frac{1}{24}, 6\frac{1}{36}, 12\frac{1}{72}$ में से सबसे छोटा:

a) $2\frac{1}{12}$ b) $3\frac{1}{18}$ c) $4\frac{1}{24}$ d) $6\frac{1}{36}$ e) $12\frac{1}{72}$

64, 81, 64, 36, 12

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③ The greatest one of $\sqrt{2}, \sqrt[3]{3}, \sqrt[6]{6}, \sqrt[5]{5}$ is

$\sqrt{2}, \sqrt[3]{3}, \sqrt[6]{6}, \sqrt[5]{5}$ में से सबसे बड़ा:

a) $\sqrt{2}$ b) $\sqrt[3]{3}$ c) $\sqrt[6]{6}$ d) $\sqrt[5]{5}$

30 × $2^{15}, 3^{10}, 6^{5}, 5^{6}$

⑥ 8, 9, 6

× ×

3^5 5^3

243 125

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④ The greatest among the numbers $\sqrt[2]{8}, \sqrt[4]{13}, \sqrt[5]{16}, \sqrt[10]{41}$ is :

$\sqrt[2]{8}, \sqrt[4]{13}, \sqrt[5]{16}, \sqrt[10]{41}$ में से सबसे बड़ी संख्या है :

a) $\sqrt[4]{13}$ b) $\sqrt[5]{16}$ c) $\sqrt[10]{41}$ d) $\sqrt[2]{8}$

$8^2, 13^1$ $16^2, 41$

✓ ✗ ✗

$8^5, 16^2$

$(2^3)^5$

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⑤ The greatest number among $3^{50}, 4^{40}, 5^{30}$ and 15^{20} is

$3^{50}, 4^{40}, 5^{30}$ और 15^{20} में से सबसे बड़ा कौनसा है?

a) 3^{50} b) 4^{40} c) 5^{30} d) 15^{20}

243, 256, 125, 225

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⑥ The greatest among $2^{550}, 3^{300}, 5^{250}, 6^{200}$ is

$2^{550}, 3^{300}, 5^{250}, 6^{200}$ में से सबसे बड़ा:

a) 2^{550} b) 3^{300} c) 5^{250} d) 6^{200}

50

2048, 729, 3125, 1296

3^3 6^2

27 ✗ 36 ✓

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(7) Find the smallest among the following

निम्न में से सबसे छोटा कौनसा है?

a) $1+3\sqrt{2}$ ~~a)~~

b) $2+\sqrt{10}$

c) $3+\sqrt{5}$

d) $4+\sqrt{3}$ ~~d)~~

18

49

$11+2\sqrt{10}$

40

10

16

5

1

6+2√5

20

$4+2\sqrt{3}$

12

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(8) Which of the following statement(s) is/are true?

निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?

I. $\sqrt{11} + \sqrt{7} < \sqrt{10} + \sqrt{8}$

II. $\sqrt{17} + \sqrt{11} > \sqrt{15} + \sqrt{13}$

a) Only I b) Only II

c) Both I and II d) Neither I nor II

$\sqrt{5}+\sqrt{4} < \sqrt{10}+\sqrt{2}$

$5+4+2\sqrt{20} < 10+2+2\sqrt{20}$

$11+7+2\sqrt{11\times7}$ $10+8+2\sqrt{10\times8}$

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(9) Which of the following statement(s) is/are TRUE?

निम्नलिखित में से कौन सा/से कथन सत्य है/हैं?

I. $\sqrt{5} + \sqrt{5} > \sqrt{7} + \sqrt{3}$

II. $\sqrt{6} + \sqrt{7} > \sqrt{8} + \sqrt{5}$

III. $\sqrt{3} + \sqrt{9} > \sqrt{6} + \sqrt{6}$

a) Only I b) Only I and II c) Only II and III d) Only I and III

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(10) Arrange the following expressions in descending order:
निम्न को घटते क्रम में लिखिए:

P) $2\sqrt{2} + \sqrt{7}$ $\sqrt{8} + \sqrt{7}$ 56

Q) $3 + \sqrt{6}$ $\sqrt{9} + \sqrt{6}$ 54

R) $\sqrt{10} + \sqrt{5}$ 50

S) $\sqrt{13} + \sqrt{2}$ 26

a) $R > Q > P > S$ b) $P > Q > S > R$

c) $P > Q > R > S$ d) $R > P > S > Q$

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$\frac{10}{\sqrt{22}+\sqrt{12}}$ ✓

$\frac{10}{\sqrt{93}+\sqrt{83}}$ ✓

(11) Arrange the following expressions in descending order:
निम्न को अवरोही क्रम में लिखिए:

P) $\sqrt{22}-\sqrt{12}$ — (2)

Q) $\sqrt{93}-\sqrt{83}$ — (4)

R) $\sqrt{15}-\sqrt{5}$ — (1)

S) $\sqrt{37}-\sqrt{27}$ — (3)

a) $R > S > P > Q$

b) $S > P > R > Q$

c) $P > R > S > Q$

d) $R > P > S > Q$

$\sqrt{1\text{crore}} - \sqrt{1\text{crore}-1}$

$\sqrt{2} - \sqrt{1}$

$1.414 - 1 = .414$

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(12) Arrange the following expressions in descending order:

निम्न को अवरोही क्रम में लिखिए:

P) $\sqrt{17} - \sqrt{15}$ — ③

Q) $2\sqrt{3} - \sqrt{10}$ — ② $\sqrt{12} - \sqrt{10}$

R) $3 - \sqrt{7}$ — ① $\sqrt{9} - \sqrt{7}$

S) $\sqrt{19} - \sqrt{17}$ — ④

a) $R > Q > P > S$

b) $S > P > R > Q$

c) $P > R > S > Q$

d) $R > P > S > Q$

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(13) **Which of the following is True**

निम्नलिखित में से कोन सा सत्य है?

a) $\sqrt[3]{11} > \sqrt[2]{7} > \sqrt[4]{45}$

b) $\sqrt{7} > \sqrt[3]{11} > \sqrt[4]{45}$

c) $\sqrt{7} > \sqrt[4]{45} > \sqrt[3]{11}$

d) $\sqrt[4]{45} > \sqrt{7} > \sqrt[3]{11}$

$11^4 \quad 7^6 \quad 45^3$

$(121)^2 \quad (343)^2 \quad (2^{10} - 31 \, 5)^2$

$(45^{1.5})^2$

$45 \times \sqrt{45}$

$6 - 7$

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65^2 17^3 12^4

65^2 $(68)^2$ $(144)^2$

$(17^{1.5})^2$

$17 \times \sqrt{17}$

4

(14) **Which of the following statements(s) is/are true?**
निम्नलिखित में से कथन/कथनों में से कोन सा/से सही है?

1. $(65)^{\frac{1}{6}} > (17)^{\frac{1}{4}} > (12)^{\frac{1}{3}}$
2. $(17)^{\frac{1}{4}} > (65)^{\frac{1}{6}} > (12)^{\frac{1}{3}}$
3. $(12)^{\frac{1}{3}} > (17)^{\frac{1}{4}} > (65)^{\frac{1}{6}}$

a) only 1
b) only 2
c) only 3
d) none of these

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(15) Which of the following is true?

निम्नलिखित में से कौन सा सत्य है?

I. $\frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt{5}}$

II. $\frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt{5}}$

III. $\frac{1}{\sqrt{5}} > \frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt[4]{29}}$

IV. $\frac{1}{\sqrt{5}} > \frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt[3]{12}}$

a) Only I b) Only II c) Only III d) Only IV

12^4 29^3 5^6

$(144)^2$ $(145)^2$ $(125)^2$

$(29\sqrt{29})^2$

$\times 5$ 6

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